

Object-Oriented Software Engineering:
Using UML, Patterns and Java, 3rd Ed.

Please answer each question below.
Please write your answer in a word document. The word document should include a photo of your face in the cover page so that I can match your name to your face.

You should draw your answers in LucidChart. Otherwise (that is, If you do not draw the diagrams), you will lose 50 points.
And then, upload your word document in the folder "Homework 1" in the Brightspace.

The due date for this homework is by 11:59pm on September 24 (Tuesday).

Important reminder: You must write your own codes and give your own answers. If you copy someone else's codes or answers, it is a serious plagiarism. You will receive zero for this assignment and disciplinary actions will be taken.

Late Policy: If you submit *m* days later than the deadline, your maximum grade will be 100*(1-0.*m*).
For example, if *m*=2, your maximum grade is 100*(1-0.2)=100*0.8=80.

Modeling with UML

Q1. Model an ATM

Draw a use case diagram for an ATM (cash terminal). The system includes two actors: a customer, who draws money from his account either on the money chip on his bank card or in cash, and a security man, who fills money into the ATM.
Use cases should include: DrawCash, LoadMoneyChip, CheckAccountBallance, FillATM. Also include the following exceptional cases: OutOfMoney, TransactionAborted (i.e., customer selected the cancel button without completing the transaction) and MoneyChipOutOfOrder.
Remark: It is possible to use inheritance between use cases!

Write the flow of events and specify all fields for the use case CheckAccountBalance that you drew before. Specify relationships.

Q2. Model a book

Draw a class diagram representing a book defined by the following statement: “A book is composed of a number of parts, which in turn are composed of a number of chapters. Chapters are composed of sections.” First, focus only on classes and associations.

Add multiplicity to the class diagram you produced.

Refine the class diagram to include the following attributes:

- Book includes a publisher, publication date, and an ISBN
- Part includes a title and a number
- Chapter includes a title, a number, and an abstract
- Section includes a title and a number

Consider the refined class diagram. Note that the Part, Chapter, and Section classes all include a title and a number attribute. Use inheritance to factor out these two attributes .

Q3. Happy Hour Message

Draw a sequence diagram for the HappyHourMessage scenario in the figure 1 below. Draw only the first five messages exchanged.

Scenario name	HappyHourMessage
Participating actor instances	bob, alice: BarFlies john: Friend
Flow of events	1. Bob and Alice are sitting in their favorite pub and it’s happy hour. They want to invite their common friend John who likes cocktails and didn’t know about the newly scheduled happy hour. Alice takes out her “mobile phone” and activates the “SMS” function. 2. Alice enters John’s cell phone number and writes the message about the happy hour into the “SMS text field”. She sends the message and waits for John’s answer. 3. John, who is still at work, is alerted by a sound of his cell phone that a new “SMS” has arrived. He reads the lines from Alice and answers that he will come immediately. He quits working and leaves the office. 4. Alice receives John’s answer at her “mobile phone”.

Figure 1

Q4. Model an online bookstore

Consider the process of ordering a book via Internet. Draw an activity diagram representing each step of the process, from the moment you open the shop's web page to the point when you start reading the book, including login or registration. Include activities that others need to perform.

Add additional exception handling to the activity diagram you have developed. Model the following three exceptions:

1. The order got lost in the Internet
2. The wrong book is packaged
3. The ordered book is no longer available